

QMEA

1. A hospital wants to determine whether the mean waiting time in its emergency department differs from 30 minutes. A random sample of 25 patients has a mean waiting time of 34 minutes. The population standard deviation is unknown.

Which test should be used?

- A. One-sample Z test
- B. One-sample t test
- C. Two-sample t test
- D. F test

2. A researcher wants to compare the mean monthly spending of customers from two independent stores. Population variances are unknown.

Which test is most appropriate?

- A. One-sample t test
- B. Independent two-sample t test
- C. Two-proportion Z test
- D. Randomized block ANOVA

3. Before comparing the means of two populations using the equal-variance approach, a researcher wants to determine whether their population variances are equal.

Which test should be used?

- A. Chi-square test
- B. F test
- C. Z test
- D. ANOVA

4. A researcher compares the mean annual salaries of employees from four different industries.

Which test should be used?

- A. Independent two-sample t test
- B. One-way ANOVA
- C. Randomized block ANOVA
- D. Chi-square test of independence

5. Ten patients each receive three different pain treatments on separate occasions. The researcher wants to compare the mean pain scores among the three treatments while controlling for differences among patients.

Which method is most appropriate?

- A. One-way ANOVA
- B. Randomized block ANOVA
- C. Independent two-sample t test
- D. Chi-square goodness-of-fit test

6. Suppose $\widehat{Salary} = 25,000 + 3,200(\text{Experience})$

where Experience is measured in years.

What does 3,200 represent?

- A. Every employee earns exactly \$3,200.
- B. An additional year of experience is associated with an estimated \$3,200 increase in salary.
- C. Experience explains 3,200% of salary variation.
- D. The average salary is \$3,200.

7. A regression model predicting income from age and education produces

$$R^2 = .1895$$

Which interpretation is correct?

- A. Age and education cause 18.95% of income.
- B. 18.95% of individuals have predictable incomes.
- C. 18.95% of the variation in income is explained by age and education in the regression model.
- D. The model predicts income correctly 18.95% of the time.

8. A political researcher surveys 800 adults and records each person's age and whether the person supports or does not support a proposed policy. The researcher wants to determine whether the proportion supporting the policy is greater than 50%.

- A. One-Proportion Z Test
- B. Two-Proportion Z Test
- C. Chi-Square Test of Independence
- D. Multiple Regression

9. A researcher uses the same survey but now wants to determine whether policy support is associated with age group (18–29, 30–49, 50–64, 65+).

- A. One-Proportion Z Test
- B. Two-Proportion Z Test
- C. Chi-Square Test of Independence
- D. One-Way ANOVA

10. A restaurant chain believes that customers choose four meal types in the proportions 30%, 25%, 25%, and 20%. A random sample of 500 purchases is used to examine this claim.

- A. One-Proportion Z Test
- B. Chi-Square Goodness-of-Fit Test
- C. Chi-Square Test of Independence
- D. One-Way ANOVA

11. A machine is designed to fill bottles with an average of 500 ml. The manufacturer knows from long-term production records that the population standard deviation is 4 ml. A sample of 50 bottles is used to determine whether the mean filling volume is less than 500 ml.

- A. One-Sample Z Test
- B. One-Sample t Test
- C. Chi-Square Variance Test
- D. One-Proportion Z Test

12. Four teaching methods are compared. Because students differ greatly in ability, students are first grouped according to similar prior GPA, and each group is exposed to the different teaching methods. The outcome is exam score.

- A. One-Way ANOVA
- B. Randomized Block ANOVA
- C. Chi-Square Test of Independence
- D. Multiple Regression

13. A researcher wants to determine whether the number of hours studied can be used to predict a student's final examination score.

- A. One-Sample t Test
- B. Simple Linear Regression
- C. Multiple Regression
- D. One-Way ANOVA

14. A researcher wants to predict examination scores using hours studied, class attendance, and prior GPA.

- A. Simple Linear Regression
- B. Multiple Regression
- C. One-Way ANOVA
- D. Randomized Block ANOVA

15. A hospital introduced a new discharge counseling program. Historically, 15% of patients were readmitted within 30 days. After implementing the program, 300 discharged patients were followed, and 33 were readmitted. The hospital wants to determine whether the new program is associated with a lower readmission rate than the historical rate.

Which statistical test is most appropriate?

- A. One-Sample t Test
- B. One-Proportion Z Test
- C. Two-Proportion Z Test
- D. Chi-Square Test of Independence

16. An agricultural researcher wants to compare the mean rice yield per rai under four fertilizer programs. Because soil quality differs substantially across locations, each of 12 farms is divided into four comparable plots, and all four fertilizer programs are applied within each farm.

Which statistical method is most appropriate?

- A. One-Way ANOVA
- B. Randomized Block ANOVA
- C. Multiple Regression
- D. Independent Two-Sample t Test

17. An online retailer classifies customers into three membership levels: Basic, Silver, and Gold. For each customer, the retailer records whether the customer's preferred payment method is credit card, bank transfer, or mobile wallet. Management wants to determine whether payment preference is related to membership level.

Which statistical method is most appropriate?

- A. One-Way ANOVA
- B. Chi-Square Goodness-of-Fit Test
- C. Chi-Square Test of Independence
- D. Two-Proportion Z Test